Generative AI in Analytics Workshop

https://github.com/lluang/GenAIForAnalytics_INFORMS2026

INFORMS Analytics+

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About me

- Lead Data Scientist - Highmark Health
- Operations research
  - Simulation, optimization, mathematical modeling
  - Machine learning/Predictive analytics
- Gen AI projects with a range of customers
  - Writers
  - Strategy
  - Finance
  - Partner relationships
  - Disaster response

Overview

- Introduction to Generative AI as an analytics aide
- Health insurance example with INFORMS Analytics Framework
- Generative AI failure modes
- Workshop example for Gen AI for use in analytics with Finance
  - You do not need to know modeling, programming, or finance to participate in the workshop
- Premise: Gen AI is useful, especially to help communication between analysts, SMEs, and decision makers. But makes errors on the details.

Introduction to Generative AI as an analytics aide

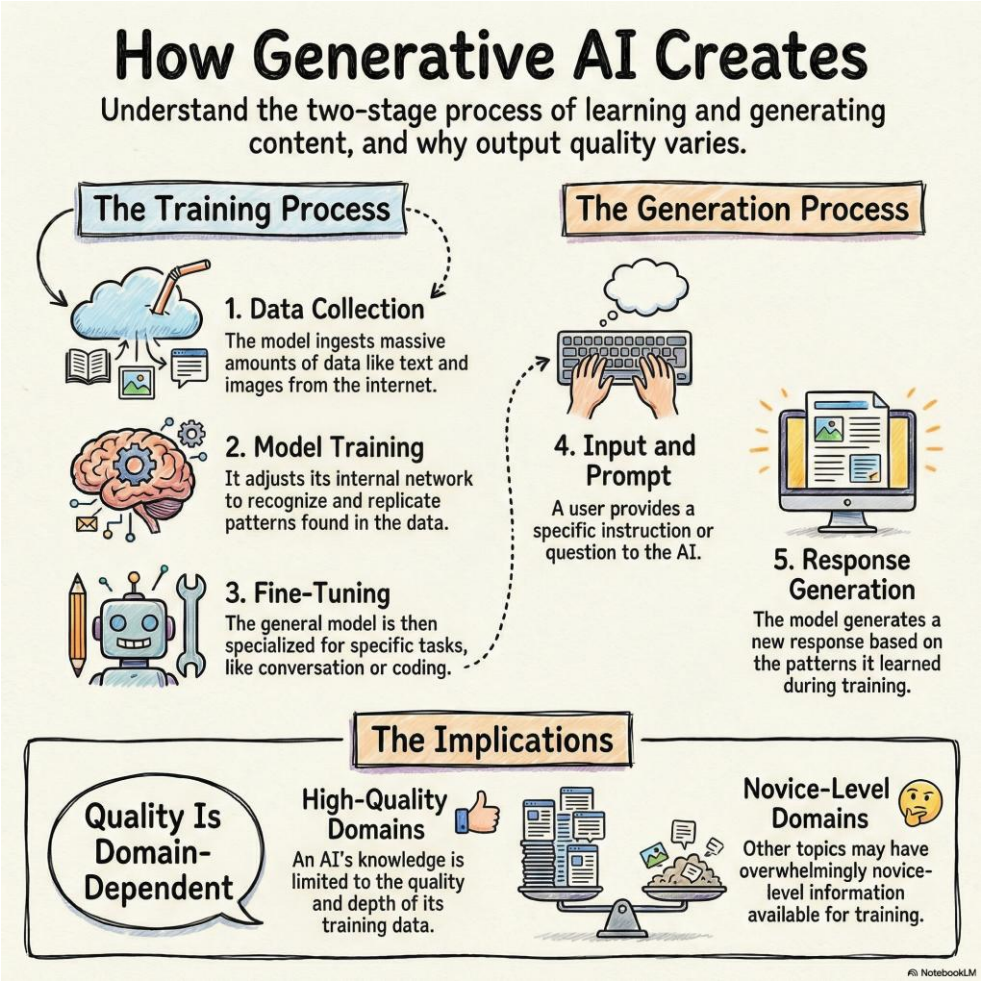
- **What is Generative AI?**
  - Definition: AI models designed to generate text, images, music, or other data from patterns learned during training.
  - Common Applications: Chatbots, content creation, image generation, music composition, computer programming.
- **How Does It Work?**
- **Why use Generative AI in analytics?**

Why use Generative AI in Analytics

- Generative AI – One year of experience in many domains
- All of the skills outside your specialty
  - Programming with data
  - Different programming environments
  - Using new/different tools, libraries, and methods
- Translation
  - Interpret information from unfamiliar knowledge domain
  - Write for an audience unlike yourself
- Always available colleague
  - Most domains claim collaboration leads to better outcomes
  - But in real life this is inconvenient
  - Generative AI is an always available colleague

How does it work

- **Training Process**
  - Foundation models built on vast amounts of data
- **Generation Process**
  - Input and Prompt: Users provide a prompt or query.
  - Response Generation: The model generates content based on patterns found in training data.
    - Random selection
  - Refinement
- **Implications**
  - Will be able to create a response
  - Quality depends on what is publicly available and published.
  - Many topics often at novice level



Gen AI Persona

Gen AI is like an overeager, naïve, new graduate hire

Overeager	Will always give an answer
Naïve	Does not know the implications of facts
New graduate	Knows a lot of facts
New hire	Knows nothing about the company or its values

Generated by Google Gemini 2024

Example problem: RAND Health Insurance Experiment

- 1982 study on cost sharing in health care and effect on service use, quality of care, and health.
- Data is readily available as a statsmodels example dataset and other sources
- Study is well described in RAND publications
    - https://www.rand.org/pubs/research_briefs/RB9174.html
- Goal: work with a specific client who needs specific insight based on the study to take specific action
    - A regional department of health exploring developing an insurance program

Business problem framing

- Goal: Understand what problem needs to be solved in order to support the activities of the business
- What makes this hard:
  - The business partner often presents the problem as a request for a product.
  - But to be useful, it should be a problem to be solved or a question to be answered.
- Danger: Failure of imagination
- Generative AI: Look at overall background and suggest topics for an initial meeting

RAND IE example prompt

- You are an analyst working with a regional health department trying to design a health insurance program.
- You have data from the RAND health insurance experiment study https://www.rand.org/pubs/research_briefs/RB9174.html
- What are some topics to bring up with the health department as possible topics of study
- Make a list of 3 topics with some idea on how potential results will be used

1. Cost-Sharing and Care Quality

The RAND study revealed that cost-sharing reduces the use of both high-value medical services and unnecessary care at the same rate. This suggests that patients generally lack the medical expertise to "shop" for only the most essential treatments when faced with out-of-pocket costs. To address this, the department can use these findings to implement **Value-Based Insurance Design**, where life-saving treatments are fully covered while discretionary procedures require a co-pay.

2. Chronic Disease and Hypertension Management

For the poorest participants, free care led to measurable improvements in blood pressure control, which directly correlates to a lower risk of stroke and heart attack. This finding highlights that for chronic conditions, removing the cost of primary care visits and medication is a life-saving investment. The department can use this to prioritize **"first-dollar coverage" for chronic disease monitoring**, potentially reducing long-term mortality rates within the region.

3. Emergency Department (ED) Utilization

The HIE demonstrated that while cost-sharing successfully reduces ED visits, it can also lead to patients delaying care until their conditions become more acute. This "blunt instrument" approach saves money on minor visits but may increase the complexity of cases that eventually arrive at the hospital. The results can be used to design a system that pairs **higher ED co-pays for non-emergencies** with free, 24/7 access to nurse hotlines or urgent care centers.

Business problem framing comments

- The Gen AI created a variety options and directions.
- Gen AI gives the answer using terms specific for that domain.
- The original version of this answer also suggested potential “how results will be used” for each.
- Helps with the “failure of imagination” problem by creating conversation hooks.

Analytical problem framing

- Goal: identify an outcome and suggest analytics course of action options
- "Here is a data set from an health insurance experiment with variable names. How should I analyze the relationship between outpatient visits and the other variables"
- Provide context
  - Description of available data
  - Data dictionary
  - The type of answer you need to make a decision

Analytics problem framing commentary

- Gen AI gave a statistical study plan
  - Warning: Gen AI tends to suggest statistics or machine learning
- It warns of potential pitfalls

Analyzing the relationship between outpatient visits (`mdvis`) and the other variables in the RAND Health Insurance Experiment data requires a multi-faceted approach, considering the nature of the variables and potential confounding factors. Here's a suggested plan:

1. Descriptive Statistics and Visualization:

* **Univariate Analysis:** Start by examining the distribution of each variable individually. Pay close attention to the distributions of `lncoins`, `lpi`, and `fmde` as they are log-transformed.

* **Bivariate Analysis:** Explore the pairwise relationships between `mdvis` and each predictor variable.

2. Regression Analysis:

Given the nature of your data, a multiple linear regression model is the most appropriate approach. The dependent variable is `mdvis` (number of outpatient visits), and the independent variables are your other variables.

3. Robustness Checks:

* **Alternative Regression Methods:** Consider using robust regression techniques (e.g., Huber-White standard errors) to account for potential heteroscedasticity (unequal variances of residuals).

* **Nonlinear Relationships:** If scatter plots suggest non-linear relationships between `mdvis` and some predictors, you might consider transforming the predictors (e.g., using polynomial terms) or using Generalized Additive Models (GAMs) which allow for flexible non-linear relationships.

Working with Data

- Working with your organization data stack
- Reading in the data
- Identifying data issues
- Common data cleaning tasks
- Data preparation/transformation into a form needed for analysis

Methodology framing

- Suggest methods that will give answers in the form that are needed
- Gen AI provides suggestions with warning
- Ask Gen AI for alternative methods of analysis

2. Regression Analysis:

Given the nature of your data, a multiple linear regression model is the most appropriate approach. The dependent variable is `mdvis` (number of outpatient visits), and the independent variables are your other variables.

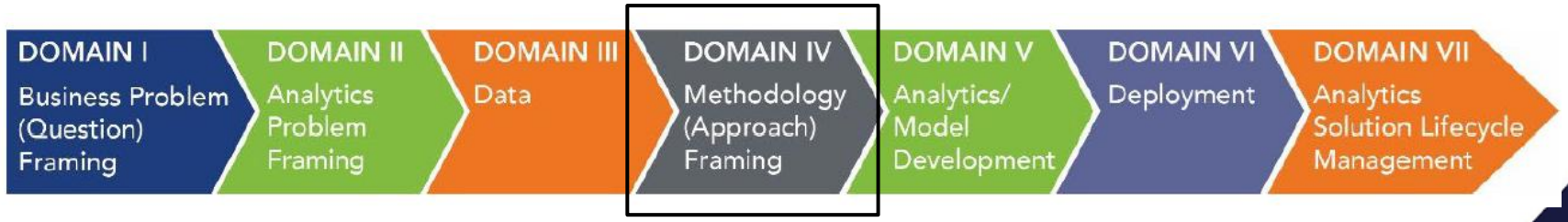
* **Log-transformed Variables:** The variables `lncoins`, `lpi`, and `fmde` are already log-transformed. Interpreting coefficients for these variables will be in terms of percentage changes in `mdvis` for a one-unit change in the *log* of the predictor.

* **Multicollinearity:** Check for multicollinearity among the predictor variables. High multicollinearity can inflate standard errors and make it difficult to interpret the individual effects of predictors.

* **Categorical Variables:** The variables `idp`, `physlm`, `hlthg`, `hlthf`, and `hlthp` are binary (0/1). These can be included directly in the regression model.

* **Model Specification:** You'll likely need to consider interaction terms. Including interaction terms (e.g., `lncoins*physlm`) allows you to test this.

* **Model Selection:** Start with a full model including all variables and interactions you deem important. Then, use model selection techniques to identify the most parsimonious model that adequately explains the data.

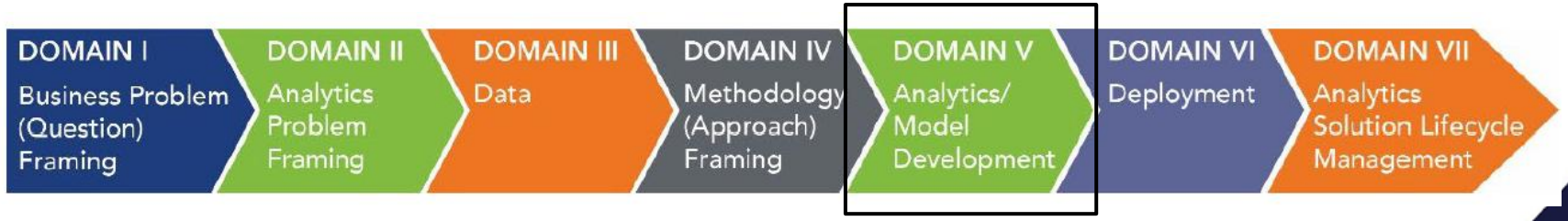


Analytics/Model development

- Ask Gen AI to implement the chosen model
- Give Gen AI examples of how to develop the model
  - Identify libraries and provide API references
  - Examples of model such as optimization or simulation textbooks that use your libraries

```
hie_ols_model = smf.ols(formula =
       'mdvis ~ lncoins + idp + lpi + fmde + physlm + disea + hlthg + hlthf +
      hlthp', data = randhie).fit()
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                   mdvis   R-squared:                       0.069
Model:                             OLS   Adj. R-squared:                  0.068
Method:                  Least Squares   F-statistic:                     165.5
Date:                 Fri, 20 Jun 2025   Prob (F-statistic):          7.54e-304
Time:                         22:02:17   Log-Likelihood:                -58316.
No. Observations:                20190   AIC:                         1.167e+05
Df Residuals:                    20180   BIC:                         1.167e+05
Df Model:                            9
Covariance Type:             nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      1.7379      0.084     20.646      0.000       1.573       1.903
lncoins       -0.1695      0.020     -8.406      0.000      -0.209      -0.130
idp           -0.7533      0.075     -9.998      0.000      -0.901      -0.606
lpi            0.1066      0.014      7.860      0.000       0.080       0.133
fmde          -0.1001      0.011     -8.707      0.000      -0.123      -0.078
physlm         1.0658      0.103     10.320      0.000       0.863       1.268
disea          0.1217      0.005     25.006      0.000       0.112       0.131
hlthg         -0.0487      0.067     -0.730      0.465      -0.179       0.082
hlthf          0.2201      0.122      1.807      0.071      -0.019       0.459
hlthp          1.4410      0.261      5.527      0.000       0.930       1.952
==============================================================================
```



Model interpretation

- Ask GenAI to interpret analytics output
- Note that it treats analytics output as a structured, so it will work through the structure
- Works through every statistic in the regression summary

Overall Model Fit:

* **R-squared (0.069):** This indicates that the model explains only 6.9% of the variance in the number of outpatient visits. This is a relatively low R-squared, suggesting that many factors influencing outpatient visits are not included in this model.

Individual Predictor Variables:

The coefficients represent the change in the average number of outpatient visits associated with a one-unit change in the predictor variable, *holding all other variables constant*. Remember:

* `lncoins`, `lpi`, and `fmde` are log-transformed. A one-unit increase in these variables represents a multiplicative change (approximately equal to a percentage change for small changes).

* **lncoins (-0.1695, p < 0.001):** A one-unit increase in the log of coinsurance (plus 1) is associated with a decrease of 0.1695 outpatient visits. Since the variable is log-transformed, this means that higher coinsurance rates (higher cost-sharing for patients) are associated with fewer outpatient visits.

* **idp (-0.7533, p < 0.001):** Individuals in the individual deductible plan (`idp` = 1) have, on average, 0.75 fewer outpatient visits compared to those not in that plan.

. . .

Model diagnostics

- Gen AI works through diagnostics because the model output is structured
- Identifies violations of assumptions
- Suggests alternative methods that handle violated assumptions

Model Diagnostics:

* **Omnibus, Skew, Kurtosis, Prob(Omnibus):** **These values suggest significant departures from normality in the residuals (errors) of the model.** This is common in count data like this and warrants consideration of alternative models (e.g., negative binomial regression, Poisson regression).

In summary: This model suggests that cost-sharing, health status (especially poor health), and having a physical limitation are strongly associated with the number of outpatient visits, while the type of insurance plan also has a significant effect. However, **the low R-squared and the diagnostic issues suggest that this is a simplified model and that other important factors need consideration**, along with potentially choosing a more appropriate statistical method to accommodate the count nature of the dependent variable.

Poisson Regression

- The Gen AI suggested Poisson regression or negative binomial regressions
- Gen AI implements Poisson regression from statsmodels

```
from statsmodels.genmod.families import Poisson
from statsmodels.stats.outliers_influence import variance_inflation_factor

# Poisson Regression Model
model_poisson = smf.glm(formula='mdvis ~ lncoins + idp + lpi + fmde + physlm +
          disea + hlthg + hlthf + hlthp',
          data=randhie, family=Poisson()).fit()

Generalized Linear Model Regression Results
==============================================================================
Dep. Variable:                  mdvis   No. Observations:                20190
Model:                            GLM   Df Residuals:                    20180
Model Family:                 Poisson   Df Model:                            9
Link Function:                    Log   Scale:                          1.0000
Method:                          IRLS   Log-Likelihood:                -62420.
Date:                Fri, 20 Jun 2025   Deviance:                       83934.
Time:                        22:02:32   Pearson chi2:                 1.27e+05
No. Iterations:                     5   Pseudo R-squ. (CS):             0.3422
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      0.7004      0.011     62.741      0.000       0.678       0.722
lncoins       -0.0525      0.003    -18.216      0.000      -0.058      -0.047
idp           -0.2471      0.011    -23.272      0.000      -0.268      -0.226
lpi            0.0353      0.002     19.302      0.000       0.032       0.039
fmde          -0.0346      0.002    -21.439      0.000      -0.038      -0.031
physlm         0.2717      0.012     22.200      0.000       0.248       0.296
disea          0.0339      0.001     60.098      0.000       0.033       0.035
hlthg         -0.0126      0.009     -1.366      0.172      -0.031       0.005
hlthf          0.0541      0.015      3.531      0.000       0.024       0.084
hlthp          0.2061      0.026      7.843      0.000       0.155       0.258
==============================================================================
```

Negative Binomal Regression

```
model_negbin = smf.glm(formula='mdvis ~ lncoins + idp + lpi + fmde +
        physlm + disea + hlthg + hlthf + hlthp',
        data=randhie,family=sm.families.NegativeBinomialP()).fit()

---------------------------------------------------------------------------
AttributeError  Traceback (most recent call last)
Cell In[14], line 2
      1 model_negbin = smf.glm(formula='mdvis ~ lncoins + idp +
        lpi + fmde + physlm + disea + hlthg + hlthf + hlthp',
----> 2 data=randhie, family=sm.families.NegativeBinomialP()).fit()

AttributeError: module 'statsmodels.genmod.families' has no
attribute 'NegativeBinomialP'
```

- Negative Binomial failed!
- Why?
- There is a ‘NegativeBinomialP’ function in statsmodels, but not in this branch of statsmodels

Generative AI: Conflations

- Generative AI works by identifying tokens in training materials that are the same or correspond with tokens in the inquiry and randomly choosing from training data to create a response.
- statsmodels has two branches that implement discrete regression models.
  - Either will work. But switching randomly between them fails because they are different API calls.
  - IRL: SMEs will choose one branch based on the type of work done and document it as a standard for the organization.

Negative Binomal Regression v2

- Correct NegativeBinomial function call
- Runs successfully

```
model_negbin = smf.glm(formula='mdvis ~ lncoins + idp + lpi + fmde + physlm +
      disea + hlthg + hlthf + hlthp',
                  data=randhie,family=sm.families.NegativeBinomial()).fit()

------------------------------------------------------------------------------
Negative Binomial Model Results:
                 Generalized Linear Model Regression Results
==============================================================================
Dep. Variable:                   mdvis   No. Observations:                20190
Model:                             GLM   Df Residuals:                    20180
Model Family:        NegativeBinomial   Df Model:                            9
Link Function:                    Log   Scale:                          1.0000
Method:                          IRLS   Log-Likelihood:                -43541.
Date:                Fri, 20 Jun 2025   Deviance:                       25175.
Time:                        22:02:52   Pearson chi2:                 3.46e+04
No. Iterations:                     7   Pseudo R-squ. (CS):            0.09843
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      0.6648      0.023     29.267      0.000       0.620       0.709
lncoins       -0.0577      0.006    -10.446      0.000      -0.069      -0.047
idp           -0.2664      0.021    -12.959      0.000      -0.307      -0.226
lpi            0.0408      0.004     11.090      0.000       0.034       0.048
fmde          -0.0379      0.003    -12.116      0.000      -0.044      -0.032
physlm         0.2687      0.027      9.964      0.000       0.216       0.322
disea          0.0380      0.001     29.534      0.000       0.035       0.041
hlthg         -0.0428      0.018     -2.361      0.018      -0.078      -0.007
hlthf          0.0198      0.032      0.609      0.542      -0.044       0.083
hlthp          0.1809      0.067      2.718      0.007       0.050       0.311
==============================================================================
/home/louis/Documents/GenAIGemini/venv/lib/python3.10/site-
packages/statsmodels/genmod/families/family.py:1367: ValueWarning: Negative
binomial dispersion parameter alpha not set. Using default value alpha=1.0.
  warnings.warn("Negative binomial dispersion parameter alpha not "
```

Model evaluation

- Ask Gen AI to compare the two models
- Interpret the results from the poisson and negative binomial regression summary based on the health insurance experiment
- Gen AI uses model diagnostics to determine that both models show significance
- Explains that negative binomial is the better model because of overdispersion.

1. Overall Model Fit

* **Poisson Regression:**
    * **Pseudo R-squared:** 0.06343. This is a pseudo-R-squared, and it has a different interpretation than the R-squared in a linear model. It provides a measure of how well the model fits compared to a null model (a model with only the intercept). This value suggests that the model explains a modest amount of variance in `mdvis`.
    * **Log-Likelihood:** -62420. The higher the log-likelihood (less negative), the better the fit of the model.
    * **LLR p-value:** 0.000. This indicates that the model, as a whole, is statistically significant. At least one of the predictors has a significant effect on the outcome.

* **Negative Binomial Regression:**
    * **Pseudo R-squared:** 0.01845. The pseudo-R-squared is lower for this model.
    * **Log-Likelihood:** -43384. This is a much higher log-likelihood than the Poisson model. Since the log-likelihood is higher, this suggests a better fit for the data.
    * **LLR p-value:** 0.000. The model is overall significant.

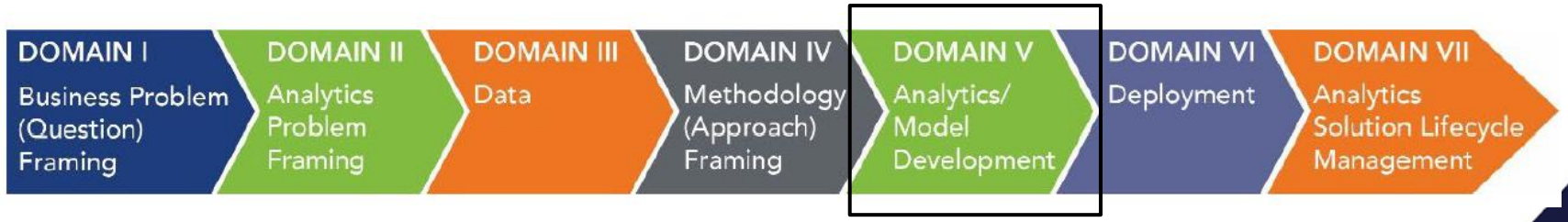
Key Differences and Model Choice:

* **Log-Likelihood:** The Negative Binomial model has a significantly *higher* log-likelihood (-43384 vs -62420). *This indicates that the Negative Binomial model provides a better fit to the data.*
* **Alpha Parameter (Negative Binomial):** The presence of the `alpha` parameter (1.2930) in the Negative Binomial model is crucial. This parameter accounts for overdispersion (the variance of the data being greater than the mean), which is common in count data. The alpha parameter is significant (p-value = 0.000), confirming overdispersion.

Conclusion: Choose the Negative Binomial model as it fits the data better.

Model development: comment

- Generative AI can be a partner in model development
- Given documentation, it is good at interpreting model diagnostics and output.
- It can make mistakes
  - Failure modes
  - Helps if it is provided references on libraries, model documentation, and model formulation



Model Deployment

- Model deployment is providing the results to the people who need it in a form that they can use to inform or improve decision making
- Software platforms: Gen AI can help transform model output into forms that the platform can deliver to the end user
- Gen AI is good at translating results into language that corresponds to the domain, leading to action

Model Interpretation

- Can ask for a summary for a specific audience.
  - Data analyst
  - Domain analyst
  - Policy analyst
  - Decision maker
  - Public communications
- Observations
  - Gen AI is more neutral with technical audiences, more optimistic with general audience.
  - Gen AI makes choices on what is important.

Okay, here's an interpretation of the Negative Binomial regression results, tailored for a public health policy analyst who may not have a strong statistical background. I'll focus on the practical implications and avoid technical jargon as much as possible.

Overall, What Did the Study Find?

This study looked at what factors affect how often people visit the doctor (outpatient visits, called `mdvis` in this study). It used data from a large health insurance experiment. The main goal was to understand how things like insurance coverage, health status, and other factors relate to doctor visits.

Key Findings (Broken Down by Variable)

Let's look at the factors that seem to be most important in predicting the number of doctor visits:

* **Insurance Costs (lncoins):** This variable relates to the amount of money people have to pay for their healthcare (called *coinsurance*).

  * **What this means:** When people have to pay more out-of-pocket for their healthcare, they tend to go to the doctor *less often*. This is a very common finding in health economics – when healthcare is more expensive, people use it less. Specifically, a one-unit increase in *lncoins* (which represents an increase in the cost of health care) is associated with a 6% decrease in outpatient visits.

* **Deductible Plans (idp):** This variable tells us if someone had a *deductible* insurance plan (where they have to pay a certain amount before insurance starts covering costs).

Lifecycle Management

- Documentation: Gen AI should be used to write/revise documentation as the model changes.
- Use Gen AI to write comments and code docstrings
- Gen AI is good for summarizing a code base. e.g. a model that has not been touched for some time.
- Ask Gen AI for a summary of changes before making any changes.

Gen AI for Programming

- Generative AI has shown itself to be successful at software engineering projects
  - Trained on public GitHub repositories
  - Many examples of use of popular programming frameworks
- But this does not always translate to analytics
  - Analytics projects are context dependent, so few useful public examples
  - Training data (GitHub) is statistics and machine learning heavy
  - Assume Gen AI is naive in analytics models and requires guidance
    - Expert review and direction
    - Model diagnostics and implications

Generative AI Failure Modes

- Novice
  - Experts think and do things differently than novices
  - Experts know to ask about what they are not told
- Naïvete
  - Facts are more well documented than implications
  - Gen AI does not recognize implications unless specified in training data
  - Training data does not usually include failures
  - Multi-domain problems. Published works typically are single domain focus. Domain experts recognize when domains intersect and how to prioritize conflicts.

Practices for using Generative AI for Analytics

- Structured prompting
- Providing technical context
- Providing methodological context
- Providing problem context

Structured Prompting - Prompt Engineering

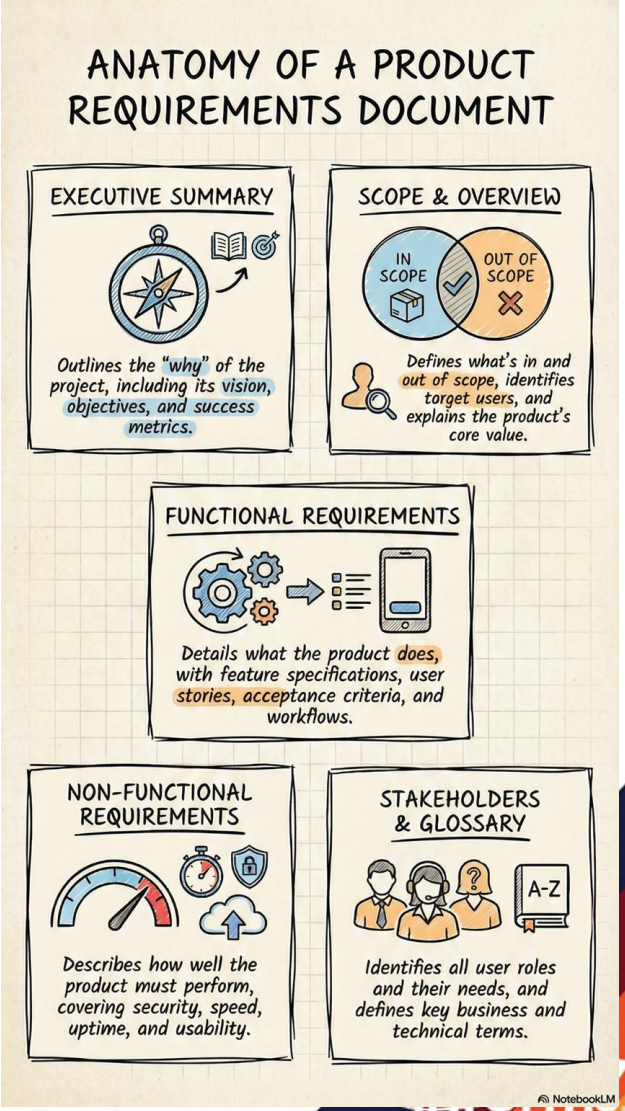
- Prompt engineering is the practice of designing prompts to enhance the quality of Generative AI
- Applies understanding of technology, language, and how people think
  - Clarity
  - Specificity
  - Thematic direction (personas)
- Can involve interaction with the Gen AI
- Prompt engineering framework: organized framework for making requests

Providing Context for Analytics

- Analytics references
  - A function reference: e.g. library API reference
  - A usage example: e.g. methods textbook
- Problem description
  - Problem statement
  - Ask generative for a mathematical formulation before asking for an implemented model
- Programming context
  - Identify frameworks and libraries you want to use
  - Enforce separation of concerns.
  - Use the mathematical formulation to check the final model

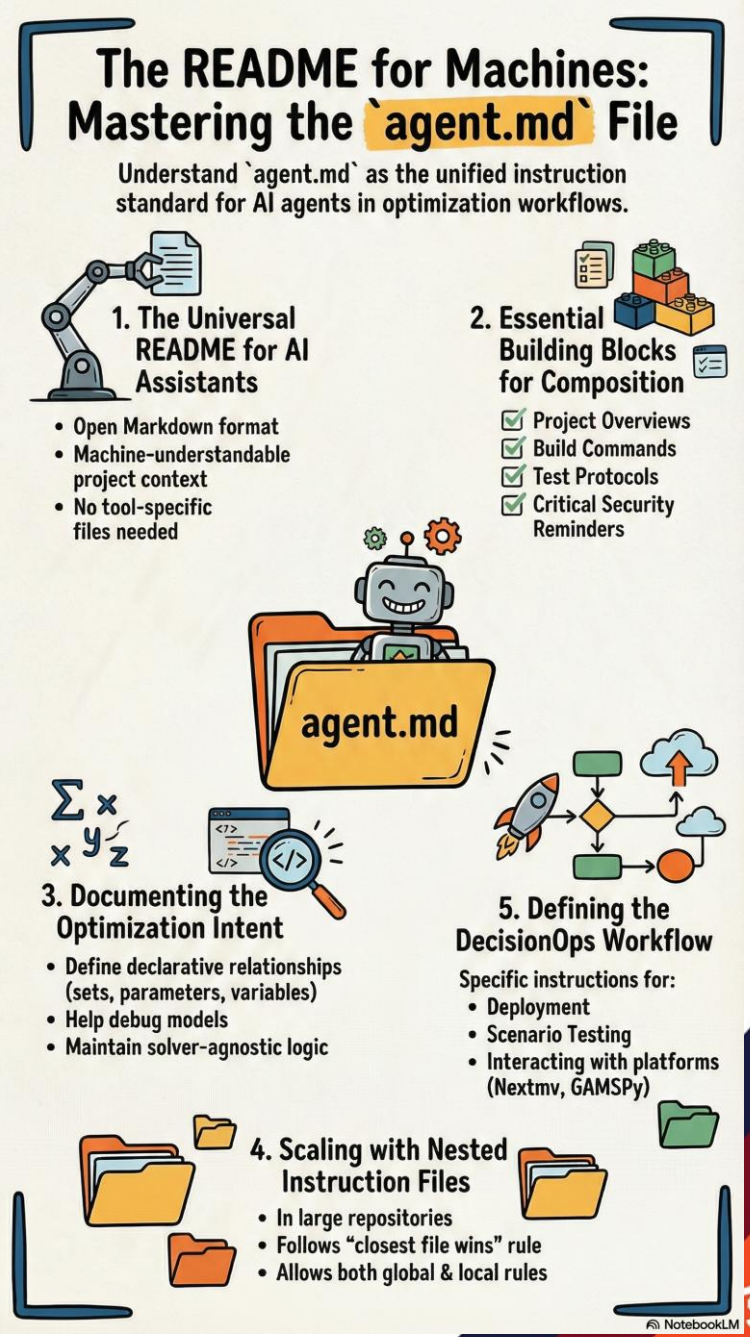
Context: Product Requirements Document

- PRD are used by project managers and product owners to specify the end product.
- Represents the understanding of the SME and the end customer (decision maker).
- Creates a common understanding of the problem and the proposed course of action.
- Language is the problem domain. Developers/analysts learn the language of the domain.



Context: Agents.md

- Standard for Gen AI terminal apps
- Project overview
- Software engineering
  - Application stack
  - Frameworks (security, UI, etc)
- Libraries and modeling frameworks
- Process
- Current state (maybe have another file for this)



Ensuring Model is Correctly Implemented

- Use Generative AI to compare:
  - Problem statement
  - Mathematical formulation
  - Model implementation
- Gen AI identifies differences; you determine what is correct
- Implement tests
- Write useful documentation
- Identify improvements for deployment
- Regularly push to a source code repository
  - Whenever you have achieved a point of progress
  - Ask for specific outcomes as you move forward

Exercise: ABC Financial

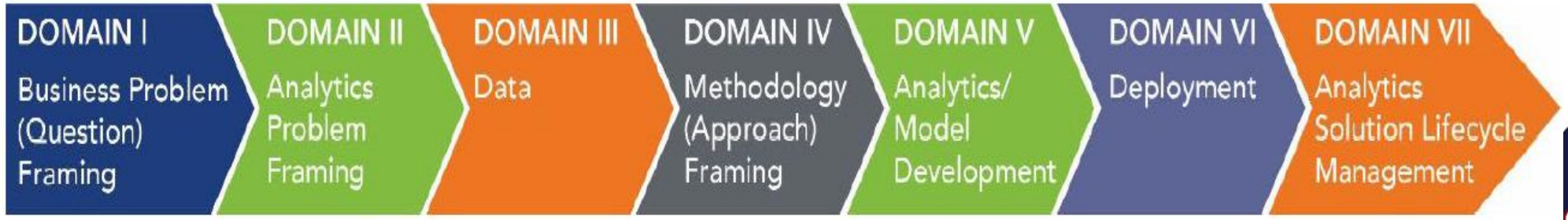
- Goal: Use Generative AI during major stages of an Analytics project lifecycle
  - Industry setting: Finance
  - Analytics method: Optimization
  - Case source: Advent of OR 2025 (FICO)
- You are NOT expected to have specific knowledge about the industry or method
  - Gen AI as a low level aide
  - Gen AI will work on problem formulation, model development, communications

Exercise setup

- Materials at github
  - https://github.com/lluang/GenAIForAnalytics_INFORMS2026
  - INFORMS Analytic Framework
  - Problem materials
  - PYOMO textbook from LLNL (https://www.osti.gov/servlets/purl/1771935)
  - PYOMO documentation (https://pyomo.readthedocs.io/_/downloads/en/stable/pdf/)
  - Gemini NotebookLM (https://notebooklm.google.com/)
    - For advanced users, use Gemini Gem, OpenAI Project, Claude Project, Copilot Agent
    - For actual model development I use terminal based tools with an IDE. Gemini CLI, Claude Code, Codex

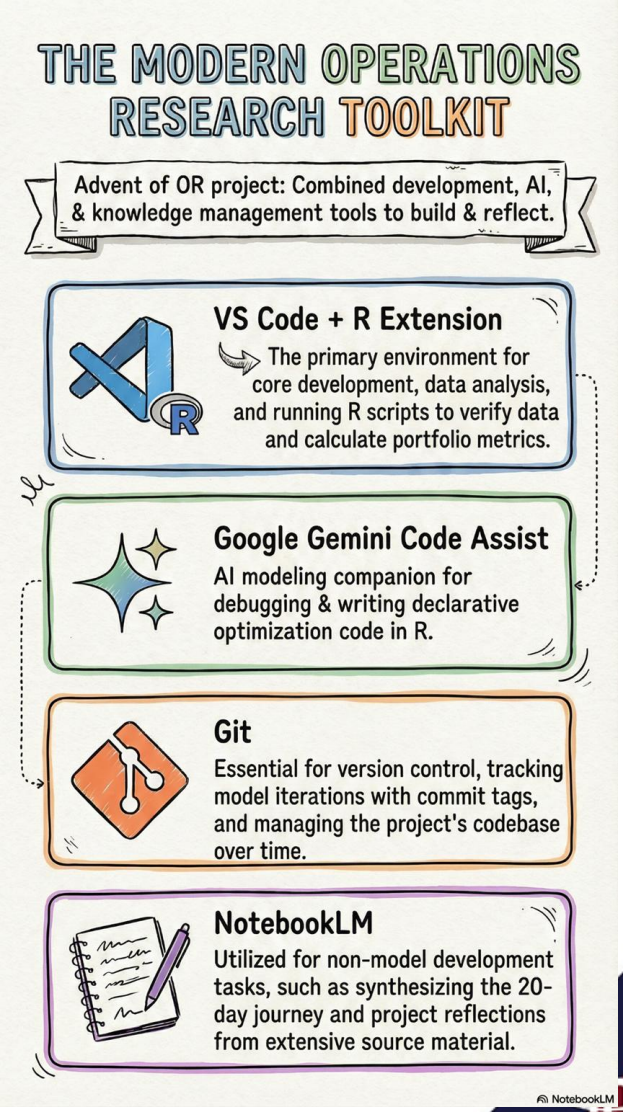
Goal of the exercise

- Goal: Use Generative AI during major stages of an analytics project lifecycle
  - Industry setting: Finance
  - Analytics method: Optimization
  - Case source: Advent of OR 2025
- Use the INFORMS Analytics Framework to define "analytics project lifecycle"



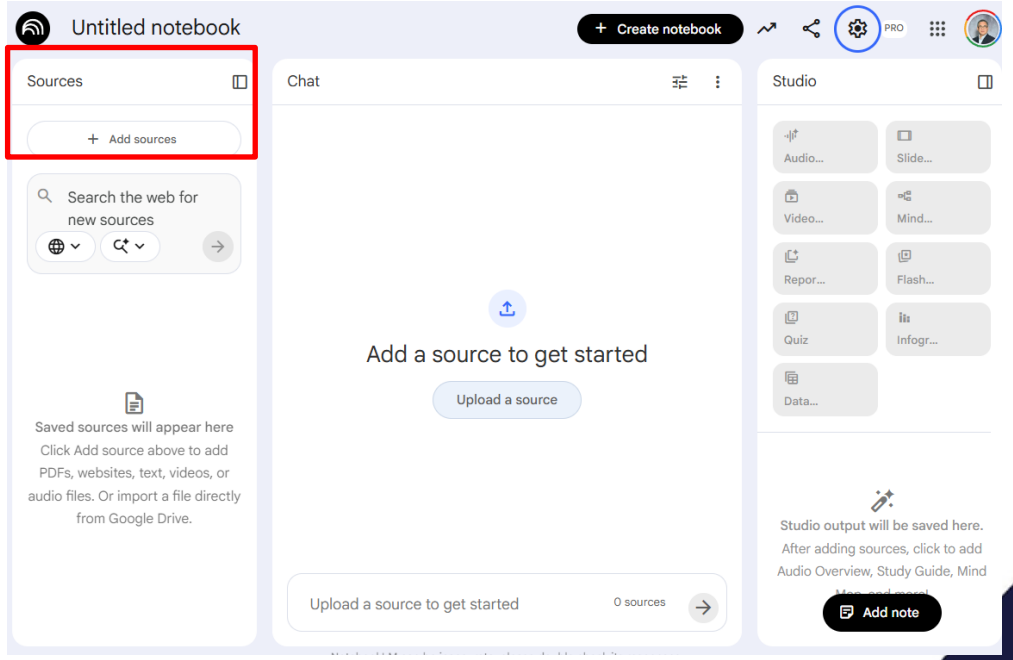
Gen AI and Analytics - Tooling

- Development environment for specifying models
  - VS Code with Gen AI extensions
- Generative AI that can work with IDE or in terminal
  - Google Gemini Code Assist (Gemini CLI)/Claude Code/OpenAI Codex/Github Copilot
- Version control to see what Gen AI changes
  - Git
- NotebookLM for non-model development tasks
  - Alternatives include Gemini Gem, ChatGPT Project, Claude Project, Copilot Project



NotebookLM setup

- Gemini NotebookLM
  - http://notebooklm.google.com
- Create new notebook
- NotebookLM is Generative AI that is based on sources you provide
  - Alternatives include Gemini Gem, ChatGPT Project, Claude Project, Copilot Project
- Github repository https://github.com/lluang/GenAIForAnalytics_INFORMS2026
- INFORMS Analytics Framework
- Problem documents



Step 1: Framing the business problem

- Copy and paste Business Problem 1.pdf Day 1 into the Notebook

The team mentioned that by regulation, the institution's total portfolio's **average risk weight** cannot exceed 50% and the total portfolio exposure cannot increase by more than 20% per quarter.

With the information you've gotten so far, think about:

Is there any additional information that you need to ask ABC financial about their problem?

What can be done to solve ABC financial's problem?

What type of analytics methodology could be used to formulate and solve the problem?

What additional information would be necessary to solve the problem?

For the next meeting you have promised to present a proposal for how to solve the problem leading you to the next step of formulating an initial analytics problem.

Business Problem Framing 2

- Copy and paste day 2 into NotebookLM

Task: Draft a brief requirements confirmation that you would send to ABC Financial after this meeting. What did you understand? What do you still need to know?

Specifically, what new decisions, objectives, constraints, data, or sets would you add?

And what questions would you ask to validate these new requirements?

Think about the data sample they promised to provide. What specific fields or columns would you expect to see? What would you check first?

Step 2: Modeling and Assumptions

- Data: upload the data files
- Copy Day 3 information (data)
- What is the structure of each table? How would you join or relate these tables?
- Given we decide quarter by quarter, and there is no forecasting, what is a reasonable way to turn historical profitability into a number to use? Would you use the mean per segment? Weighted averages? The last quarter only? The worst-case quarter as a conservative assumption? How would you justify that choice to ABC's stakeholders?
- Do you spot any inconsistencies in the names or structure of the profitability files that would require a cleaning step before merging?
- Given this data structure and trying to think a bit ahead, what type of analytics approach would make sense? A deterministic optimization using mean values? Stochastic optimization using return distributions? Multi-period optimization?
- Write python code to import these data files as pandas data frames.

Step 3: Data wrangling

- Copy Day 4

Task 1: Verify the average profitability calculation

The average_profitability column in segments.csv was calculated from the raw quarterly data.

Your job: Using loan_profitability_per_quarter.csv and segment_profitability_means.csv, try to recreate how the averages were computed. Do the numbers match? What methodology was used?

Task 2: Calculate current portfolio metrics

Using the data:

What's the total portfolio exposure?

What's the current weighted-average risk weight? (Hint: weight by exposure)

Is the 50% regulatory constraint already binding?

Task 3: Draft a mathematical formulation

In words or informal notation, write out:

What are the decision variables?

What's the objective function?

What constraints do we need?

Don't worry about perfect notation -we'll start formalizing it tomorrow. The goal is to start translating the business problem into optimization terms.

Task 4: Flag any data concerns

Is there anything in the consolidated data that you'd want to clarify with ABC Financial before building the model? (Think about the negative profitability segments, the extreme InvestmentMortgage values, etc.)

Step 4: Methodology selection

Upload the Pyomo textbook and documentation

Day 5 Methodology selection

Inventory ABC Financial's problem type

Based on what we've built so far, classify ABC Financial's portfolio model:

What type of variables do we have? (Continuous? Integer? Binary?)

Is our objective function linear or nonlinear?

Are our constraints linear or nonlinear?

What problem class does this put us in? (LP, MIP, QP, MIQP, NLP, MINLP?)

Mathematical formulation

Day 6 Mathematical formulation

Define the Decision Variables: State clearly the variable(s) that your model will solve for.

Formulate the Objective Function: Write down the mathematical expression (using summation notation) for maximizing profit net of transaction costs.

Formulate Key Constraints: Write down the mathematical expressions for at least three critical constraints.

List Model Assumptions: Detail at least three key assumptions you made to complete this initial mathematical formulation.

- Save the result (Save to note)

Pyomo implementation

Day 7 Pyomo implementation

How to implement ABC Financial's business problem.

Identify the mathematical structure of the portfolio problem and write down:

Sets.

Parameters.

Decision variables.

Objective(s).

Constraints.

Implement the model as Python Pyomo code

- Save to note

Comparing formulations

- **Day 8 Comparing formulations**
- Compare the model described in *Optimal Loan Portfolio Rebalancing and Risk Weighting Linear Model* to the Mathematical formulation and pyomo implementation found in *Financial Portfolio Optimization with Pyomo Implementation*. Check each constraint and the objective to verify if they are the same.
- Note: replace the italicized names with the names of the saved notes/files where **you saved the results earlier**
- **There should be some commentary about differences due to making slightly different assumptions in the Pyomo code. Do you agree with the changes?**

Step 5: Deployment and iteration

- I ran this model and the output is in *abcfinancial.*pdf
  - Full disclosure, I edited the Gen AI generated code and model before running it.
- Look at *abcfinancial.pdf*
- Is it understandable?
- Use Generative AI to present this in a usable form.

Present the solution

- Load *abcfinancial.pdf* into the NotebookLM
- **Day 9 present results**
- Prompt: from the abcfinancial.pdf, present the solution to the portfolio optimization model
- Prompt: present the solution given in abcfinancial.pdf to rebalance the portfolio as a set of buy and sell instructions

Step 6: Measuring impact and communication

- Gen AI is good for communicating with stakeholders
- Quantitative analysts
- Domain analysts
- Account representatives
- Management
- Customers
- Public

Interpreting results

- Note: in real life, I would upload company communications such as prospectus and investment philosophy statements to provide the Gen AI with phrasing before this part
- **Day 10 Interpret results**
- Ask for a narrative interpretation to be given to a trader
- Ask for a summary to be provided to a financial analyst who will then review the model and instructions
- Ask for a report that will be given to the investor account representative to prepare for a meeting with a client who holds this portfolio
- Ask for a memo to be given to a client who holds this portfolio to justify the suggested actions

Evaluating the outputs

- Review the results
- Are they factually the same?
- In what ways are the presentations of results different?
- If this was your work product to be given to a stakeholder, would you use this or how would you change it?

Exercise review

- We used Gen AI to:
  - take a setting and developed questions to be brought to stakeholders as part of understanding the business problem and choosing an analytic approach
  - write code to read in data
  - formulate a model to achieve the desired results
  - implement the model in a modeling language
  - confirm that the narrative, mathematical, and implemented models matched
  - analyze results
  - prepare presentations of the results to a range of audiences

Summary

- Gen AI can be a valuable aide.
  - Working with new domains, methods, or tools.
  - Converting between domain language and modeling language.
  - Developing model descriptions from unstructured problem description.
- But it makes certain types of mistakes that users should be careful of
  - Novice/nuance
  - Naivete
- We need to be providing it with context, tone, and standard processes.
- We check results and take responsibility for its use.

Acknowledgements

- Thanks to the Advent of OR and FICO for the exercise ABC Financial problem and materials
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